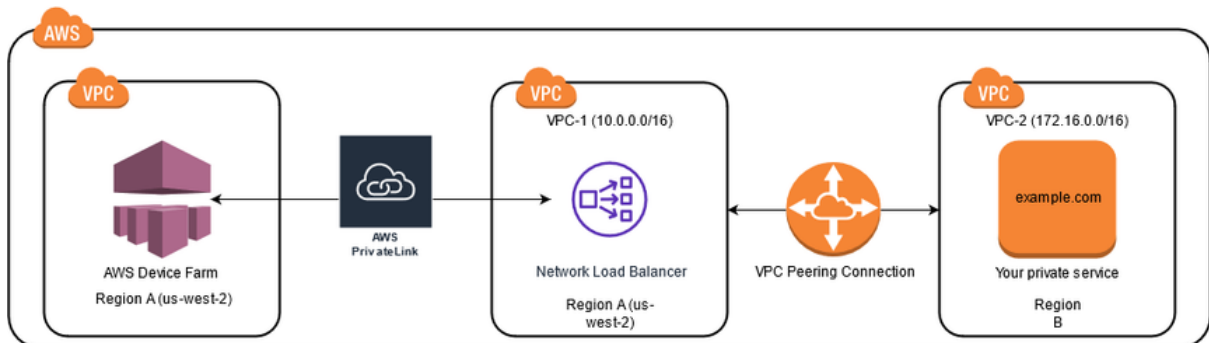


AWS Regions, Availability Zones and VPC-41

Introduction

- Amazon Web Services (AWS) is a global cloud provider with infrastructure spread across the world.
- To ensure high availability, scalability, and fault tolerance, AWS divides its infrastructure into **Regions**, **Availability Zones (AZs)**, and **Virtual Private Clouds (VPCs)**.
- These components form the backbone of how applications are deployed and managed in AWS.



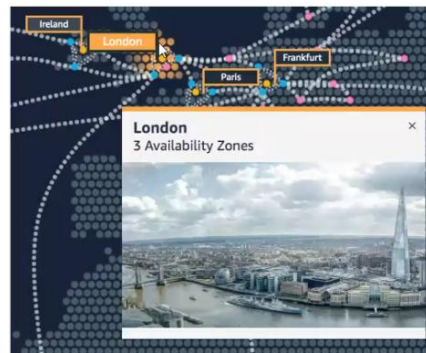
AWS Regions

- A **Region** is a physical geographic location where AWS has multiple data centers.
- Each Region consists of multiple **Availability Zones**.
- Regions allow customers to deploy resources close to their users to reduce latency.
- Data stored in one Region is not replicated automatically to another Region (unless configured).
- Regions are independent of each other for fault isolation.
- Examples of Regions:
 - US East (N. Virginia) – us-east-1
 - Asia Pacific (Mumbai) – ap-south-1
 - Europe (Ireland) – eu-west-1
- When choosing a Region, consider:

- Latency (distance to users)
- Data residency requirements
- Service availability (not all AWS services are in every Region)
- Pricing differences between Regions

AWS Regions

- An **AWS Region** is a geographical area.
 - **Data replication** across Regions is controlled by you.
 - **Communication** between Regions uses AWS backbone network infrastructure.
- Each Region provides full redundancy and connectivity to the network.
- A Region typically consists of two or more **Availability Zones**.



Example: London Region

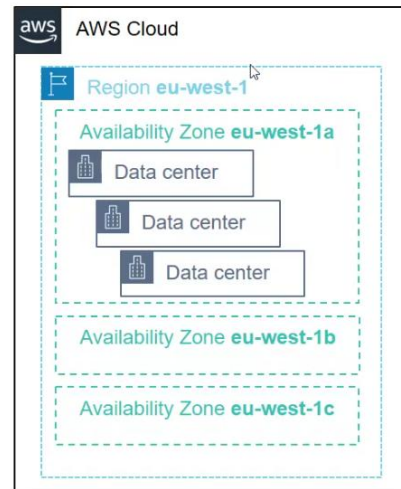


AWS Availability Zones (AZs)

- An **Availability Zone (AZ)** is one or more discrete data centers within a Region.
- Each AZ has independent power, cooling, and networking.
- AZs are physically separate but connected with **low-latency links**.
- A Region typically has **2–6 AZs** (e.g., us-east-1a, us-east-1b, us-east-1c).
- Purpose of AZs:
 - Fault tolerance: if one AZ fails, workloads can failover to another.
 - High availability: distributing resources across AZs prevents downtime.
 - Disaster recovery: multi-AZ deployment ensures business continuity.
- Examples:
 - Deploying an EC2 instance in multiple AZs for redundancy.
 - Running RDS databases with Multi-AZ replication for automatic failover.

Availability Zones

- Each **Region** has multiple Availability Zones.
- Each **Availability Zone** is a fully isolated partition of the AWS infrastructure.
 - Availability Zones consist of discrete **data centers**
 - They are designed for fault isolation
 - They are interconnected with other Availability Zones by using high-speed private networking
 - You choose your Availability Zones.
 - **AWS recommends replicating data and resources across Availability Zones** for resiliency.



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9

AWS Virtual Private Cloud (VPC)

- A **Virtual Private Cloud (VPC)** is a logically isolated virtual network within AWS.
- You define IP ranges, subnets, routing, and security policies.
- A VPC allows you to design and control your networking environment.
- Each AWS account comes with a **default VPC** in every Region, but you can create custom VPCs.

Amazon VPC



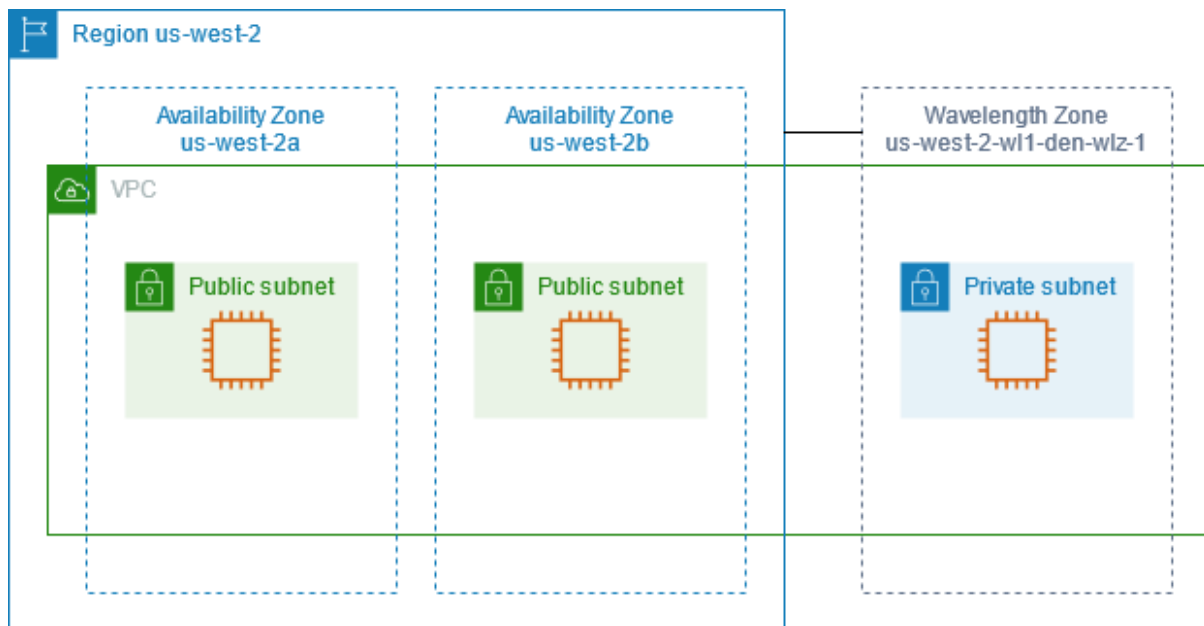
Amazon
VPC

- Enables you to provision a **logically isolated** section of the AWS Cloud where you can launch AWS resources in a virtual network that you define
- Gives you **control over your virtual networking resources**, including:
 - Selection of IP address range
 - Creation of subnets
 - Configuration of route tables and network gateways
- Enables you to **customize the network configuration** for your VPC
- Enables you to use **multiple layers of security**



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11



Key Components of a VPC

- **Subnets**
 - Divide your VPC into smaller networks.
 - Two types:
 - Public subnets (connected to the internet via Internet Gateway).
 - Private subnets (internal communication, no direct internet access).
- **Internet Gateway (IGW)**
 - Allows communication between your VPC and the internet.
- **NAT Gateway / NAT Instance**
 - Lets private subnet resources access the internet securely.
- **Route Tables**
 - Define traffic flow within the VPC and outside.
- **Security Groups (SGs)**
 - Act as virtual firewalls at the instance level.
 - Control inbound and outbound traffic.
- **Network Access Control Lists (NACLs)**

- Act as stateless firewalls at the subnet level.
- Provide an extra layer of security.
- **Peering and Transit Gateway**
 - Connect multiple VPCs within or across Regions.

Example Deployment in VPC

- Create a VPC with CIDR block 10.0.0.0/16.
- Divide into subnets:
 - 10.0.1.0/24 – Public Subnet (EC2 web server).
 - 10.0.2.0/24 – Private Subnet (Database).
- Attach Internet Gateway to VPC.
- Configure Route Table:
 - Public Subnet routes internet traffic via IGW.
 - Private Subnet routes through NAT Gateway.
- Use Security Groups:
 - Web server allows HTTP/HTTPS inbound from internet.
 - Database allows inbound only from web server's private IP.

Benefits of AWS Regions, AZs, and VPCs

- High availability and fault tolerance by spreading workloads across AZs.
- Scalability by deploying in multiple Regions.
- Security and control through VPC customization.
- Compliance with data residency laws by choosing appropriate Regions.
- Flexibility to design enterprise-level networks in the cloud.

Regions provide global coverage, **Availability Zones** ensure resilience, and **VPCs** give secure networking environments.

Together, they form the foundation of AWS cloud infrastructure, enabling businesses to build highly available, secure, and scalable applications.